

SIMATS ENGINEERING
ASSESSMENT TOOL III – ERROR ANALYSIS TASK (LOW)

Course Name:	Code:ITA1404	CO: CO2
Duration: 90 min Date:	Total Marks:	Reg. No / Name :192311053

CO2: *Execute footprinting in social engineering such as DNS analysis, competitive intelligence gathering, and attack simulations to collect relevant information for identifying potential security risks.(BL3)*
Assessment Tool Weightage: 35%

Instructions: Answer all questions. Each question carries 10 marks.

1. A student performs footprinting using Google dorks but constructs queries without operators like “site:”, “filetype:”, or quotation marks. As a result, the search results include unrelated websites and irrelevant documents. The student proceeds with analysis using this incorrect data. Identify the error in query formulation and explain why improper search syntax leads to inaccurate footprinting. Demonstrate how to correct the search query using proper operators and refine results for accurate intelligence gathering.
2. While conducting competitive intelligence, a student gathers company data from blogs and unofficial forums without verifying authenticity. The collected information includes outdated employee details and incorrect organizational structure. Identify the mistake in source validation and explain why unreliable data affects decision-making. Recommend methods to verify information using official sources such as company websites, LinkedIn, and government databases.
3. A student directly starts active scanning on a target organization without performing passive footprinting such as WHOIS lookup and search engine analysis. This results in detection alerts being triggered. Identify the mistake in reconnaissance methodology and explain why passive reconnaissance is critical. Suggest the correct sequence of footprinting steps.
4. A student performs WHOIS lookup but only records domain name and registrar details, ignoring expiry date, administrative contacts, and name servers. Identify the missing components and explain how these fields provide useful security insights. Suggest how proper WHOIS analysis can reveal vulnerabilities.
5. A student downloads publicly available documents from an organization’s website but focuses only on the visible content while ignoring embedded metadata. As a result, important details such as author names, software versions, and internal usernames are missed. Identify the mistake in the student’s footprinting process and explain how metadata can provide valuable intelligence. Describe the correct method to extract metadata using appropriate tools. Suggest how this information could be used ethically to identify potential security risks.

Code of Conduct:

I S Venkata Abhishek(192311053)(Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:Abhishek

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Error Identification	20%				
Root Cause Analysis	30%				
Correction & Justification	25%				
Application of Concepts	15%				
Clarity & Presentation	10%				

Criteria	Error Identification (2)	Root Cause Analysis (3)	Correction & Justification (2)	Application of Concepts (2)	Clarity & Presentation (1)	Total
Q1						
Q2						
Q3						
Q4						
Q5						

Q1. Incorrect Google Dork Query Formulation

Error Identification

The student's mistake is **improper search-query formulation**. The student used broad search terms without search operators such as `site:`, `filetype:`, and quotation marks. Therefore, the search engine returned unrelated websites and documents.

Root Cause Analysis

Search operators are used to restrict and refine search results. Without them, the search engine searches for individual keywords across a large number of websites. This produces irrelevant results and may cause the student to collect incorrect information during footprinting.

For example, a general search such as:

`employee directory example company`

may return information from many unrelated organizations.

The student may then incorrectly assume that those results belong to the target organization.

Correction

The student should use appropriate search operators. For an authorized security assessment, examples include:

`site:example.com employee`

This restricts results to the specified domain.

To identify publicly indexed PDF documents:

`site:example.com filetype:pdf`

To search for an exact phrase:

`site:example.com "employee directory"`

These operators reduce irrelevant results and improve the accuracy of information gathering.

Security Significance

Proper search syntax helps identify relevant publicly exposed information such as documents, employee information, and technical material while reducing false results.

Conclusion

The error occurred because the student used **unrestricted search queries**. Using `site:`, `filetype:`, and quotation marks produces more focused and accurate footprinting results.

Q2. Failure to Validate Competitive Intelligence Sources

Error Identification

The student collected information from **unofficial blogs and forums without verifying its authenticity**. As a result, outdated employee information and an incorrect organizational structure were used in the analysis.

Root Cause Analysis

The main problem is **poor source validation**.

Information on unofficial websites may be:

Outdated.

Incorrect.

Incomplete.

Biased.

Published without verification.

Using such information in competitive intelligence can lead to incorrect conclusions and poor security decisions.

For example, an outdated employee profile could cause the analyst to incorrectly identify an employee as a current decision-maker.

Correction

Information should be cross-checked using reliable sources such as:

- **Official company website** — verify company structure, leadership, services, and contact information.
- **LinkedIn** — verify current professional roles and organizational relationships.
- **Government databases** — verify legally registered company information.
- **Multiple independent sources** — compare information before accepting it.
- **Recent publications** — check whether the information is current.

Security Significance

Accurate intelligence helps security teams correctly identify:

1. Organizational structure.
2. Important personnel.
3. Technology used by the organization.
4. Public-facing assets.
5. Potential security risks.

Incorrect intelligence can cause the organization to focus on the wrong people, systems, or threats.

Conclusion

Competitive intelligence must be based on **reliable and verified sources**. Official company websites, LinkedIn, government databases, and multiple sources should be used to validate information.

Q3. Starting Active Scanning Without Passive Footprinting

Error Identification

The student directly started **active scanning** without first performing passive reconnaissance such as WHOIS lookup and search-engine analysis. This caused detection alerts to be triggered.

Root Cause Analysis

The mistake is an **incorrect reconnaissance methodology**.

Active scanning directly interacts with the target infrastructure. Such activity can generate:

Firewall alerts.

IDS/IPS alerts.

Security monitoring events.

Suspicious network traffic.

Passive reconnaissance, on the other hand, gathers publicly available information without directly probing the target.

Passive reconnaissance helps the analyst understand the target before deciding what active techniques are necessary.

Correct Sequence

The appropriate sequence is:

1. Define scope and authorization
- ↓
2. Passive footprinting
- ↓
3. WHOIS / RDAP analysis
- ↓
4. Search-engine reconnaissance
- ↓
5. DNS analysis
- ↓
6. Identify domains, subdomains and IP ranges
- ↓
7. Analyze the attack surface
- ↓
8. Authorized active scanning
- ↓
9. Analyze findings
- ↓
10. Report security risks

Why Passive Reconnaissance Is Critical

Passive reconnaissance:

- Reduces unnecessary interaction with the target.
- Helps avoid premature detection.
- Provides initial attack-surface information.
- Makes active testing more focused.
- Helps prevent unnecessary scanning of irrelevant systems.

Conclusion

The student should **perform passive reconnaissance first and active scanning only afterward within the authorized scope**. This produces a safer and more systematic security assessment.

Q4. Incomplete WHOIS Analysis

Error Identification

The student recorded only the **domain name and registrar** while ignoring other useful WHOIS fields such as:

- Expiry date.
- Administrative contacts.
- Name servers.

The assessment specifically identifies these missing components.

Root Cause Analysis

WHOIS information contains several pieces of information that can contribute to understanding a domain's infrastructure and ownership.

By recording only the domain and registrar, the student loses important contextual information.

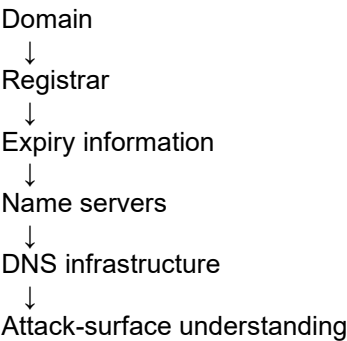
Missing Components and Their Importance

WHOIS Field	Security/Reconnaissance Value
Domain name	Identifies the target domain
Registrar	Identifies the organization responsible for registration services
Expiry date	Shows registration lifecycle information
Administrative contact	May provide organizational/contact information where publicly available
Name servers	Helps identify the DNS infrastructure supporting the domain

Correction

The student should record all relevant publicly available WHOIS/RDAP information and then correlate it with DNS and other passive reconnaissance results.

For example:



Security Significance

Proper WHOIS analysis can help identify:

Domain ownership information.

DNS infrastructure.

Organizational contacts.

Related infrastructure.

Domains requiring further authorized assessment.

It can also reveal outdated or unexpected registration information that should be reviewed by the security team.

Conclusion

The mistake was performing **incomplete WHOIS analysis**. A complete analysis should consider all relevant registration and DNS-related fields rather than only the domain name and registrar.

Q5. Ignoring Document Metadata

Error Identification

The student downloaded publicly available documents but examined only their visible content. The student failed to analyze **embedded metadata**, resulting in important information such as author names, software versions, and internal usernames being missed.

Root Cause Analysis

Documents can contain metadata that is not immediately visible when reading the document.

Metadata may include:

- Author name.
- Organization name.
- Creation date.
- Modification date.
- Software/application used to create the document.
- Software version.
- Internal usernames.
- Document properties.
- Other technical information.

Therefore, analyzing only the visible document content gives an incomplete picture of the information exposed by the organization.

Correct Method

The analyst should perform metadata extraction on documents obtained from an **authorized assessment**.

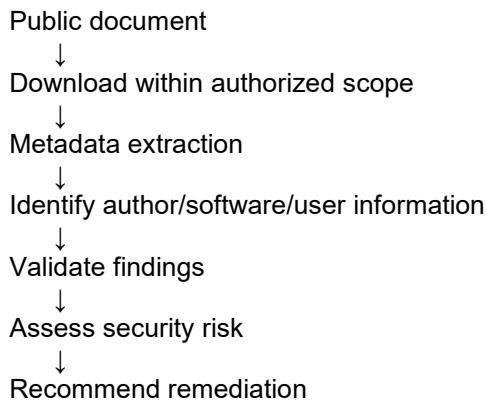
Suitable tools include:

ExifTool — extracts metadata from many file formats.

FOCA — can assist with metadata and document information gathering.

Built-in document properties for formats such as Microsoft Office documents.

A typical workflow is:



Security Significance

Metadata can reveal information that helps attackers understand an organization's environment.

For example:

Author name → **Employee identification**

Software version → **Technology identification**

Internal username → **Account naming information**

This information could potentially support social engineering or further reconnaissance.

Ethical Use

The information should be used only to:

Identify accidental information exposure.

Assess the organization's security posture.

Recommend removal of unnecessary metadata.

Improve document-handling procedures.

It should **not** be used to target employees or systems without authorization.

Conclusion

The student's mistake was ignoring document metadata. Proper metadata extraction provides additional intelligence and can help organizations identify and remove unnecessary information exposure.